

Transient Shaper

A real-time envelope shaper implemented as a JUCE VST3 effect for Windows x64.

DSP overview

The processor separates fast onsets from slower signal body using two independent one-pole peak envelopes for each audio channel. A sensitivity-scaled difference between the envelopes becomes a transient weight. That weight blends the Attack and Sustain gain controls in decibels. A short gain smoother then reduces abrupt control changes before the shaped gain and final Output trim are applied. The algorithm is amplitude-envelope based; it is not a compressor with a threshold and ratio, nor a frequency-dependent processor.

Processing flow



For each sample, the fast envelope uses 0.8 ms attack and 55 ms release; the body envelope uses 18 ms attack and 135 ms release. The target gain is smoothed with a 3.5 ms time constant. These time constants are fixed in the current implementation and are recalculated as coefficients for the active sample rate.

Core equations

$e[n] = \alpha * e[n-1] + (1 - \alpha) * \text{abs}(x[n])$; α selects attack or release according to level direction.

$T[n] = \text{clamp}(((eFast[n] - eBody[n]) / (eFast[n] + 1e-5)) * 2.5 * \text{Sensitivity}, 0, 1)$

$G_{\text{target}}[n] = \text{Attack_dB} * T[n] + \text{Sustain_dB} * (1 - T[n])$

$y[n] = x[n] * \text{gain}(G_{\text{smoothed_dB}}[n]) * \text{gain}(\text{Output_dB})$

The transient weight is bounded from 0 to 1. At zero Attack and Sustain, the shaper stage is neutral; Output remains an independent post-shaper gain.

Technical Summary

DSP overview, processing flow, and implementation decisions | Version 0.0.1

Implementation decisions

Area	Implementation
Parameters	JUCE AudioProcessorValueTreeState; Attack and Sustain: -12 to +12 dB; Sensitivity: 0.25x to 4.0x; Output: -12 to +12 dB. Defaults are 0 dB, 0 dB, 1.0x and 0 dB.
Real-time path	Cached parameter atomics and fixed two-channel envelope arrays are used in processBlock. The block loop performs no explicit dynamic allocation. Gain and meter values are read and written through atomics.
Channel support	The main input/output layout accepts matching mono or stereo buses. Envelope and gain states are held independently per channel.
State and automation	Parameters are automatable and saved/restored through the APVTS XML state. pluginval exercised parameter automation and state restoration.
Meters and display	Input and output peak dB plus average transient activity are published atomically. The custom 650 x 420 editor refreshes its meter/display at 24 Hz. The envelope graphic is a parameter illustration with a live detector marker, not a waveform display.
Latency and tail	The processor reports 0 samples of latency and 0 seconds of tail. There is no lookahead, oversampling, or frequency-selective sidechain in this version.

Build and validation

The project uses C++17 and JUCE, building Windows VST3 and Standalone targets through CMake. The VST3 reports the name Transient Shaper, manufacturer Archie DSP, version 0.0.1 and category Fx | Dynamics. Build and plugin validation results are documented separately in the Testing Report.

Scope note: Attack and Sustain are gain amounts in dB. Sensitivity changes how much the fast-versus-body envelope difference contributes to the transient weight; it is not a threshold or attack-time control.

Source and build files: <https://github.com/amitabhsarkarwya/transient> (commit e3a6e81).